

ENGINEERING

# ServicePro Delivery Roadmap

## 1. Current State

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# Document Control

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| <b>Purpose</b>  | Enterprise platform overview and buyer evaluation reference                                 |
| <b>Audience</b> | Business leaders, platform administrators, evaluators, partners, and technical stakeholders |
| <b>Scope</b>    | Capabilities, architecture, security, deployment, adoption, outcomes, and terminology       |
| <b>Source</b>   | ServicePro repository documentation; technical meaning preserved                            |

Version: 8.0.0-alpha.1 Date: July 2026 Planning Horizon: 6 months

ServicePro is a functional alpha with strong feature coverage for field service operations. The platform serves real tenants (Aqua Pro Plumbing, C & D Landscaping) but runs on free-tier infrastructure without production hardening. The gap between "working code" and "production-grade platform" is the focus of this roadmap.

## 2. Workstream Overview

| # | Workstream             | Priority | Timeline    | Objective                               |
|---|------------------------|----------|-------------|-----------------------------------------|
| 1 | Production Hardening   | Critical | Weeks 1–4   | Eliminate service reliability risks     |
| 2 | Developer Experience   | High     | Weeks 2–6   | CI/CD, testing, type safety             |
| 3 | Frontend Modernization | High     | Weeks 4–10  | Performance, maintainability, UX        |
| 4 | API Architecture       | Medium   | Weeks 6–14  | Framework adoption, router refactor     |
| 5 | Data Platform          | Medium   | Weeks 8–16  | Backups, monitoring, connection pooling |
| 6 | Observability          | Medium   | Weeks 10–18 | Error tracking, metrics, alerting       |
| 7 | Security Hardening     | High     | Weeks 4–12  | Scanning, rotation, penetration testing |
| 8 | Feature Completion     | Ongoing  | Weeks 1–24  | Customer portal, mobile, admin UI       |

## 3. Workstream Details

### 3.1 Production Hardening (Weeks 1–4)

Goal: Make the platform reliable enough for paying customers.

| Task                                                   | Effort  | Impact                        |
|--------------------------------------------------------|---------|-------------------------------|
| Upgrade Render API to Starter plan (\$7/mo)            | 1 hour  | Eliminates 50s cold starts    |
| Upgrade Supabase to Pro (\$25/mo)                      | 1 hour  | Daily backups, no pause, 8 GB |
| Add health check alerting (UptimeRobot or similar)     | 2 hours | Know when service is down     |
| Configure proper CORS for production domain only       | 1 hour  | Tighten security surface      |
| Add package-lock.json to root and enforce npm ci       | 1 hour  | Reproducible builds           |
| Move JWT_SECRET to environment variable (not generate) | 1 hour  | Survives redeploy             |
| Document recovery procedure for DB restoration         | 4 hours | Runbook for incidents         |

Exit Criteria: API responds in <2s on first request, database has daily backups, alerting notifies on downtime.

## 3.2 Developer Experience (Weeks 2–6)

Goal: Prevent regressions, enforce quality, enable safe collaboration.

| Task                                             | Effort  | Impact                          |
|--------------------------------------------------|---------|---------------------------------|
| Add TypeScript --noEmit check to CI              | 2 hours | Catch type errors before deploy |
| Run test suite in CI (even if some fail, report) | 4 hours | Visibility into test health     |
| Add ESLint with recommended rules                | 4 hours | Code consistency                |
| Enable branch protection on main                 | 30 min  | Require PR for changes          |
| Add PR template with checklist                   | 1 hour  | Structured review               |
| Set up Playwright for E2E smoke tests            | 8 hours | Critical path verification      |
| Create .env.local.example for onboarding         | 1 hour  | New developer setup             |
| Document local development setup in README       | 2 hours | Reduce onboarding friction      |

Exit Criteria: CI blocks PRs with type errors, at least 5 E2E tests covering login/jobs/storefront, branch protection enabled.

## 3.3 Frontend Modernization (Weeks 4–10)

Goal: Improve performance, reduce component complexity, establish design system.

| Task                                                       | Effort   | Impact                |
|------------------------------------------------------------|----------|-----------------------|
| Break StorefrontBuilder into 5–8 sub-components            | 12 hours | Maintainability       |
| Break PublicStorefront into section components             | 8 hours  | Testability           |
| Extract shared UI primitives (Button, Input, Modal, Card)  | 16 hours | Consistency           |
| Remove CSS !important overrides — use proper specificity   | 8 hours  | Predictable styling   |
| Add component-level Storybook or similar catalog           | 8 hours  | Visual regression     |
| Implement proper loading/error states for all data fetches | 8 hours  | UX polish             |
| Add skeleton screens for slow API responses                | 4 hours  | Perceived performance |
| Audit accessibility (WCAG 2.1 AA) for core flows           | 8 hours  | Compliance            |

Exit Criteria: No component exceeds 400 lines, shared UI library covers 80% of patterns, no !important in stylesheets.

## 3.4 API Architecture (Weeks 6–14)

Goal: Replace fragile raw HTTP routing with maintainable framework.

| Task                                                      | Effort   | Impact            |
|-----------------------------------------------------------|----------|-------------------|
| Evaluate Express vs. Fastify (recommend Fastify for perf) | 4 hours  | Informed decision |
| Extract route groups from router.js into modular files    | 16 hours | Maintainability   |
| Adopt chosen framework incrementally (adapter pattern)    | 24 hours | Modern middleware |
| Remove or lazy-load Phase 9–45 stub modules               | 8 hours  | Faster startup    |
| Standardize error response format across all routes       | 8 hours  | API consistency   |

| Task                                         | Effort   | Impact        |
|----------------------------------------------|----------|---------------|
| Generate OpenAPI spec from route definitions | 8 hours  | Documentation |
| Add request/response validation (Zod or Joi) | 16 hours | Input safety  |

Exit Criteria: Router.js under 200 lines, all routes registered via framework router, startup time under 3 seconds.

### 3.5 Data Platform (Weeks 8–16)

Goal: Production-grade database operations.

| Task                                                        | Effort  | Impact                    |
|-------------------------------------------------------------|---------|---------------------------|
| Implement connection pooling (Supabase Vibes or PgBouncer)  | 4 hours | Connection efficiency     |
| Add slow query logging (pg_stat_statements)                 | 4 hours | Performance visibility    |
| Set up automated backup verification (restore to temp)      | 8 hours | Backup confidence         |
| Add database migration CI check (dry-run)                   | 4 hours | Prevent broken migrations |
| Implement proper migration versioning (sequential numbered) | 8 hours | Schema history            |
| Add tenant data isolation tests                             | 8 hours | Security verification     |
| Plan upgrade path from free tier when needed                | 2 hours | Capacity planning         |

Exit Criteria: Connection pooling active, slow queries logged, backup restore verified monthly.

### 3.6 Observability (Weeks 10–18)

Goal: See what's happening in production without SSH access.

| Task                                                    | Effort  | Impact                 |
|---------------------------------------------------------|---------|------------------------|
| Add Sentry for error tracking (free tier: 5K events/mo) | 4 hours | Error visibility       |
| Add structured JSON logging                             | 8 hours | Log searchability      |
| Implement request duration tracking                     | 4 hours | Performance baseline   |
| Add database query timing                               | 4 hours | Bottleneck detection   |
| Set up Render metrics dashboard                         | 2 hours | Resource monitoring    |
| Add custom health metrics (tenant count, job volume)    | 4 hours | Business metrics       |
| Configure alerting thresholds                           | 4 hours | Proactive notification |

Exit Criteria: All unhandled errors captured, P95 latency visible, alerts for error spike and downtime.

### 3.7 Security Hardening (Weeks 4–12)

Goal: Close security gaps before acquiring more customers.

| Task                                              | Effort  | Impact                     |
|---------------------------------------------------|---------|----------------------------|
| Add npm audit to CI with failure on high severity | 2 hours | Dependency safety          |
| Implement JWT secret rotation mechanism           | 8 hours | Secret hygiene             |
| Add CSRF protection for state-changing endpoints  | 8 hours | Request forgery prevention |

| Task                                                 | Effort   | Impact               |
|------------------------------------------------------|----------|----------------------|
| Implement rate limiting per tenant (not just global) | 4 hours  | Abuse prevention     |
| Add input sanitization audit for all user inputs     | 8 hours  | Injection prevention |
| Enable Supabase Row Level Security for shared tables | 12 hours | Defense in depth     |
| Schedule quarterly dependency update review          | 2 hours  | Maintenance cadence  |
| Document security incident response procedure        | 4 hours  | Readiness            |

Exit Criteria: Zero high-severity npm audit findings, per-tenant rate limiting active, RLS on shared tables.

### 3.8 Feature Completion (Ongoing)

Goal: Complete customer-facing features that drive revenue.

| Feature                                                          | Effort   | Priority |
|------------------------------------------------------------------|----------|----------|
| Customer Portal — full self-service (booking, payments, history) | 40 hours | High     |
| Admin UI — complete platform management interface                | 24 hours | High     |
| Mobile responsive improvements (remaining pages)                 | 16 hours | Medium   |
| Email notifications (transactional via SendGrid/Resend)          | 12 hours | High     |
| Payment processing integration (Stripe)                          | 20 hours | High     |
| SMS notifications (Twilio)                                       | 8 hours  | Medium   |
| Reporting export to PDF/Excel                                    | 8 hours  | Medium   |
| Multi-user support per workspace                                 | 12 hours | Medium   |

## 4. Resource Requirements

### 4.1 Infrastructure Costs (Monthly)

| Service            | Current | Recommended      | Cost    |
|--------------------|---------|------------------|---------|
| Render API         | Free    | Starter          | \$7/mo  |
| Render DB          | Free    | — (use Supabase) | \$0     |
| Supabase           | Free    | Pro              | \$25/mo |
| Cloudflare Workers | Free    | Free             | \$0     |
| Sentry             | —       | Free tier        | \$0     |
| UptimeRobot        | —       | Free tier        | \$0     |
| Total              | \$0     | —                | \$32/mo |

### 4.2 Scaling Thresholds

| Trigger              | Action                               | Cost Impact |
|----------------------|--------------------------------------|-------------|
| >5 tenants active    | Upgrade Render to Standard (\$25/mo) | +\$18/mo    |
| >50 concurrent users | Add Redis caching                    | +\$15/mo    |
| >1 GB database       | Supabase already Pro                 | Included    |

| Trigger      | Action                      | Cost Impact |
|--------------|-----------------------------|-------------|
| >100 tenants | Evaluate multi-instance API | +\$50/mo    |

## 5. Milestone Schedule

| Milestone               | Target  | Key Deliverables                                |
|-------------------------|---------|-------------------------------------------------|
| M1: Stable Production   | Week 4  | Paid hosting, backups, alerting, no cold starts |
| M2: CI/CD Complete      | Week 6  | Tests in CI, branch protection, type checking   |
| M3: Security Baseline   | Week 12 | Audit clean, RLS, rate limiting, rotation       |
| M4: Framework Migration | Week 14 | Fastify/Express adopted, router refactored      |
| M5: Frontend V2         | Week 10 | Component library, no overrides, accessible     |
| M6: Observability       | Week 18 | Full error tracking, metrics, alerting          |
| M7: Feature Complete    | Week 24 | Customer portal, payments, email, mobile        |

## 6. Risk Register

| Risk                                          | Likelihood | Impact   | Mitigation                         |
|-----------------------------------------------|------------|----------|------------------------------------|
| Free tier service outage during customer demo | High       | High     | Upgrade to paid plans (M1)         |
| Database pause loses customer data            | Medium     | Critical | Supabase Pro + backup verification |
| Regression deployed to production             | High       | Medium   | CI test gating (M2)                |
| Security breach via unvalidated input         | Medium     | Critical | Input validation audit (M3)        |
| Developer confusion from 14K file count       | High       | Low      | Documentation, .gitignore cleanup  |
| Router.js becomes unmaintainable              | High       | Medium   | Framework migration (M4)           |
| Customer-visible errors with no alerting      | High       | Medium   | Sentry integration (M6)            |

## 7. Decision Log

| Decision                              | Rationale                                    | Date      |
|---------------------------------------|----------------------------------------------|-----------|
| Stay on Supabase (not Render DB)      | Better free tier, familiar, pooling built-in | July 2026 |
| Static export for frontend            | Cloudflare Workers = fast, free, global CDN  | July 2026 |
| No Express/Fastify yet                | Working system, refactor after stabilization | July 2026 |
| Phase 9–45 remain scaffolded          | Future roadmap, don't delete, but lazy-load  | July 2026 |
| Monorepo stays (no microservices yet) | Single deploy target, simple, 2 tenants      | July 2026 |